

Market Impact Analysis

```
import pandas as pd

df = pd.read_csv('market_impact.csv')
```

```
print(df.info())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 358 entries, 0 to 357
Data columns (total 31 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   incident_id                          358 non-null    str
 1   stock_ticker                         358 non-null    str
 2   price_7d_before                      358 non-null    float64
 3   price_disclosure_day                 358 non-null    float64
 4   price_1d_after                      358 non-null    float64
 5   price_7d_after                      358 non-null    float64
 6   price_30d_after                     358 non-null    float64
 7   volume_avg_30d_baseline              358 non-null    int64
 8   volume_disclosure_day                358 non-null    int64
 9   sector_index                        358 non-null    str
10   sector_return_same_period            358 non-null    float64
11   abnormal_return_1d                  358 non-null    float64
12   abnormal_return_7d                  358 non-null    float64
13   abnormal_return_30d                 358 non-null    float64
14   car_negl_to_pos1                    358 non-null    float64
15   car_0_to_7                          358 non-null    float64
16   car_0_to_30                         358 non-null    float64
17   car_0_to_90                         358 non-null    float64
18   t_statistic_1d                      358 non-null    float64
19   p_value_1d                          358 non-null    float64
20   t_statistic_30d                     358 non-null    float64
21   p_value_30d                         358 non-null    float64
22   earnings_announcement_within_7d     358 non-null    bool
23   market_cap_at_disclosure            358 non-null    float64
24   volume_ratio_disclosure             358 non-null    float64
25   pre_incident_volatility_30d         358 non-null    float64
26   post_incident_volatility_30d        358 non-null    float64
27   days_to_price_recovery              322 non-null    float64
28   notes                               92 non-null     str
29   created_at                          358 non-null    str
30   updated_at                          358 non-null    str
dtypes: bool(1), float64(22), int64(2), str(6)
memory usage: 84.4 KB
None
```

Analysis

```
# Market reaction summary
market_reaction = df[[
    "abnormal_return_1d",
    "abnormal_return_7d",
    "abnormal_return_30d",
    "car_negl_to_pos1",
    "car_0_to_7",
    "car_0_to_30",
    "car_0_to_90"
]]

print(market_reaction.describe())
```

	abnormal_return_1d	abnormal_return_7d	abnormal_return_30d	\
count	358.000000	358.000000	358.000000	
mean	-0.034183	-0.030746	-0.017018	
std	0.022088	0.026497	0.024767	
min	-0.098128	-0.106112	-0.107945	
25%	-0.049536	-0.046946	-0.031301	
50%	-0.031195	-0.028201	-0.017532	
75%	-0.018867	-0.010302	0.000988	
max	0.009984	0.025921	0.054253	

	car_negl_to_pos1	car_0_to_7	car_0_to_30	car_0_to_90
count	358.000000	358.000000	358.000000	358.000000
mean	-0.035832	-0.030746	-0.017018	-0.012242
std	0.023884	0.026497	0.024767	0.021135
min	-0.110026	-0.106112	-0.107945	-0.095579
25%	-0.052601	-0.046946	-0.031301	-0.024614
50%	-0.032509	-0.028201	-0.017532	-0.010039
75%	-0.017551	-0.010302	0.000988	0.003258
max	0.012444	0.025921	0.054253	0.038986

```
# Trading Volume Summary
volume_summary = df[[
    "volume_avg_30d_baseline",
    "volume_disclosure_day",
    "volume_ratio_disclosure"
]]

print(volume_summary.describe())
```

	volume_avg_30d_baseline	volume_disclosure_day	volume_ratio_disclosure
count	3.580000e+02	3.580000e+02	358.000000
mean	3.556234e+06	9.897671e+06	2.746851
std	7.377916e+06	2.063455e+07	0.722342
min	2.746300e+04	8.053000e+04	1.508100
25%	4.439210e+05	1.188034e+06	2.114125
50%	1.166098e+06	3.107902e+06	2.753400
75%	2.963769e+06	7.480774e+06	3.351200
max	6.688953e+07	1.728628e+08	3.993300

```
# Volatility Summary
volatility_summary = df[[
    "pre_incident_volatility_30d",
    "post_incident_volatility_30d"
]]

print(volatility_summary.describe())
```

	pre_incident_volatility_30d	post_incident_volatility_30d
count	358.000000	358.000000
mean	0.023778	0.037085
std	0.008535	0.014843
min	0.010135	0.011879
25%	0.016342	0.024839
50%	0.023608	0.034480
75%	0.030760	0.047426
max	0.039934	0.077072

```
# Recovery Summary
recovery_summary = df["days_to_price_recovery"]
print(recovery_summary.describe())
```

count	322.000000
mean	109.866460
std	103.606842
min	5.000000
25%	28.000000
50%	57.000000
75%	177.000000
max	363.000000

Name: days_to_price_recovery, dtype: float64

Start coding or [generate](#) with AI.